ALSU ALIYEVA

Data Science | Machine Learning | AI
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PROFESSIONAL SUMMARY

Third-year Computer Engineering student at Azerbaijan University of Economics (UNEC), focused on data science and applied AI. Works across data analysis, visualization and workflow automation with Python, SQL, Power BI, Tableau and n8n, and has built projects in deep learning, satellite imagery analysis, retrieval-augmented generation and automated threat detection. Won first place at a national digital ideas competition, and built teamwork and problem-solving skills through hackathons and technical competitions. Also mentors students in Python and data science.

EDUCATION

Azerbaijan University of Economics (UNEC) **Expected Jun 2028**
Bachelor of Science in Computer Engineering Baku, Azerbaijan

- cGPA 3.34/4.00 | DIM entrance score 615/700

Matrix Academy
Data Science Program

EXPERIENCE

Data Scientist Intern **Jun 2026 – Sep 2026**
QSS Analytics Baku, Azerbaijan (On-site)

- Cleaned and validated datasets for statistical analysis, checking them for accuracy and consistency.
- Analyzed data for patterns and trends and turned the findings into insights for business decisions.

Python Mentor **Jun 2026 – Sep 2026**
Data Science Academy Baku, Azerbaijan (On-site)

- Mentored a cohort of 12 students in Python, SQL and machine learning during on-site practical sessions.
- Graded 100+ assignments, quizzes and exams and helped each student work on their weak areas.

PROJECTS

Brain Tumor Segmentation **Sep 2026 – Oct 2026**
PyTorch, U-Net, FastAPI, Grad-CAM

- Built an end-to-end deep learning pipeline in PyTorch using a U-Net architecture to segment brain tumors across 3,000+ MRI scans, achieving a 0.745 Dice score and real-time inference via FastAPI. Integrated Grad-CAM for explainable AI predictions, providing clinicians with interpretable visual overlays alongside high-precision diagnostic support.

Enterprise RAG Knowledge Assistant **Aug 2026 – Sep 2026**
FastAPI, LangChain, FAISS, Docker, GitHub Actions

- Built a production-ready async RAG microservice with FastAPI, LangChain, and FAISS, serving context-aware docs with <100ms vector search latency via Docker and GitHub Actions CI/CD.

UrbanScan AI **Jun 2026 – Sep 2026**
Random Forest, Markov Chain, Streamlit, Web GIS

- Classified land cover across 3,000+ ha using Random Forest on high-resolution satellite imagery (88.5% accuracy, 0.83 Kappa), engineered a 4-year Markov Chain urban growth model, and analyzed WHO green space metrics via a custom Streamlit Web GIS platform.

Superstore Sales Analysis **Apr 2026 – May 2026**
Python, KMeans, RFM, Plotly, Seaborn

- Analyzed 9,900+ sales records using KMeans clustering and RFM metrics to segment customers into 4 behavioral groups, isolated top 10 loss-making accounts, and pinpointed severe margin decline driven by aggressive discounts (>20%) via interactive Plotly & Seaborn dashboards.

AI Cybersecurity Risk Pipeline **Apr 2026**
n8n, Gemini LLM, JavaScript, Google Sheets

- Built a real-time cybersecurity pipeline in n8n integrated with Gemini LLM, enforcing system prompt constraints to evaluate text inputs on a 0-10 numerical risk score; classified threats into 3 distinct verdict categories (Təhlükəsiz, Şübhəli, Təhlükəli) and structured raw outputs into Google Sheets via custom JavaScript nodes.

SKILLS

- **Programming:** Python, HTML, CSS, JavaScript, Git/GitHub
- **Databases:** Oracle SQL, MySQL
- **Data Analysis & BI:** Power BI, Tableau, Microsoft Excel
- **Machine Learning & AI:** PyTorch, U-Net, Random Forest, KMeans, LangChain, FAISS, FastAPI
- **Automation:** n8n Workflow Automation, Claude, Prompt Engineering, AI-Assisted Prototyping
- **Soft Skills:** Teamwork, Problem-Solving, Critical Thinking, Project Development, Adaptability

ACHIEVEMENTS, CERTIFICATES & LANGUAGES

- **Achievement:** 1st Place, National Digital Idea Competition (Rəqəmsal İdeya Müsabiqəsi) – KidCity Quest, Kid Safety Simulation Platform
- **Activities:** Volunteer, Eurasian Conference on Human-Computer Interaction (HCI-E), Dec 2025; Digital Security Ambassador, World AzeYouth Union, Aug 2026 – Oct 2026; Volunteer, RIIB
- **Certificates:** Aptis General English Test (C1), British Council; SQL for Data Science, UC Davis (Coursera); Fundamentals of Visualization with Tableau, UC Davis (Coursera); Data Analytics Job Simulation, Deloitte; AI Skills 4 Women, Founderz; SQL (Basic) & SQL (Intermediate), HackerRank; Introduction to Data Science, Cisco; Machine Learning with Python, IBM; Machine Learning, DeepLearning.AI & Stanford
- **Languages:** Azerbaijani (Native), English (Fluent), Turkish (Fluent), Russian (Intermediate)